

to the price span in which the instrument was traded for one day based on the high, low, and close values. The formula for the true range for day t is:

$$TR_t = \max(H, C_{t-1}) - \min(L, C_{t-1})$$

The lowest price of today's low and yesterday's close is subtracted from the highest of today's high and yesterday's close, and we arrive at the range in which we have been trading for today. To get to the average true range (ATR), you just need to take an average of these numbers. I use an exponential moving average to arrive at the normal trading range for each instrument over time. You may see other smoothing methods for the ATR in other books, but it usually matters very little which one you use.

The point with using the ATR as volatility estimation is that it provides not only a comparable number that we can use it to standardise position sizes, but also as an estimation of how large a move we can expect on any given normal day. We can use this information to set a target impact of the position on the portfolio as a whole. This target impact number will in turn act as a kind of leverage factor which could be scaled up or down to shift the risk level of the strategy up or down.

For our initial strategies we use a risk factor of 20 basis points, so the theoretical average daily impact on the portfolio for each position should be 0.2%. Imagine the current portfolio as a whole is worth US\$1 million. Since we want each position to have a theoretical impact of 20 basis points, that translates to US\$2,000. We get a buy signal in gold and the ATR of this instrument happens to be exactly US\$10 at that time, and because we know that the Comex Gold contract has a point value of 100, each contract we buy has a theoretical impact of US\$1,000. So we buy two of them.

The reason I keep using the phrase 'theoretical impact' is that you must always be aware that volatility is not stationary and can change dramatically over the course of a position's lifetime. This position-sizing method does not depend on a stationary volatility, it just uses the ATR at the time of position entry as a reasonable estimation of potential price moves. The position size is kept constant through the life of the position and not increased or reduced over time. The last part is something you may or may not want to tweak in your final strategy.

The formula employed here is:

$$Contracts = \frac{0.022 * Equity}{(ATR_{100} * Point Value)}$$